Rishi Raj

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-oriented Computer Vision, Generative AI, and full-stack applications. Experienced in developing scalable AI systems using FastAPI, Python, YOLO, TensorRT, LangChain, and modern cloud-native tooling. Demonstrated expertise in multi-camera video analytics, person re-identification, real-time inference pipelines, REST API development, database design, and containerized deployments. Strong foundation in software engineering principles including modular architecture, dependency injection, CI/CD workflows, test-driven development, and performance optimization. Passionate about designing intelligent systems that bridge AI research with real-world enterprise applications.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), C++
Frontend: React.js, Next.js,Tailwind CSS, HeadlessUI, Google Stitch
Backend: FastAPI ,Node.js, RestAPI , Convex
AI / ML: TensorFlow, PyTorch, YOLO, OpenCV, TensorRT, CUDA, cuDNN, LLM Integration, Agentic AI, RAG, Embedding
Database: PostgreSQL, MySQL, MongoDB, SQL, SQLAlchemy (ORM), Prisma , Qdrant
Cloud & DevOps: AWS, Azure (AZ-900 Certified), Docker, Kubernetes, CI/CD Pipelines, Linux, Bash, SDLC.
Tools: Git, GitHub, Vercel, Render, Supabase, Prisma, Label Studio, Postman, VS Code

PROFESSIONAL EXPERIENCE

AI Engineer — *June 2026 – Present*

Uniconverge Technologies Pvt. Ltd. | Noida Sector 62 | On-site

- Developed an AI-powered crack detection system for Indian Railway suspension springs using UV imaging, Python, OpenCV, and Computer Vision, automating inspection of 1000+ spring components.
- Built a real-time defect monitoring and alert platform, reducing manual inspection time by up to 70% and improving maintenance efficiency.
- Developed an AI-based Railway Inspection System capable of detecting 25+ critical coach and bogie defects using Vision, Thermal, and Edge AI technologies.
- Built an automated defect detection pipeline processing trains at speeds up to 100 km/h, reducing manual inspection effort and improving safety monitoring.

Computer Vision / AI Engineer Intern — *March 2026 – June 2026*

Uniconverge Technologies Pvt. Ltd. | Noida Sector 62 | On-site

- Jetsen Orion Nano: Designed and developed a multi-camera object detection application processing 4 simultaneous feeds — applied software engineering best practices including modular pipeline design, performance optimization, and deployment-ready structuring using NVIDIA TensorRT and YOLO.
- MIRA Platform: Developed and integrated full-stack application features including REST API endpoints, real-time data processing, and enterprise system integration using FastAPI and SQLAlchemy; applied dependency injection patterns and test-driven development principles throughout.
- Person ReID System: Built and containerized a cross-camera identity tracking application using Docker; implemented efficient data access and transaction management via embedding-based similarity search with ResNet and YOLO-Pose; utilized CI/CD workflows for deployment.

KEY PROJECTS

MIRA – AI Mock Interview Platform [link]

Tech: *React 19, FastAPI, LangChain, LangGraph, Langfuse , Groq, MediaPipe, MySQL, JWT , RapidAPI, Sentry*

- Designed and developed a full-stack enterprise web application with real-time interview orchestration and adaptive question generation; applied dependency injection, modular service architecture, and REST API design patterns using FastAPI and LangChain.
- Integrated enterprise system components including JWT-based authentication, tokenized subscription management, resume parsing, and ATS-style skill extraction; implemented browser-side proctoring using TensorFlow.js with real-time multi-person detection.
- Designed scalable data access layer with SQLAlchemy and MySQL applying transaction management best practices; built analytics modules with performance trend visualization and domain-wise tracking; optimized frontend with lazy loading for improved application responsiveness.

Silo – AI-Powered Full-Stack Application Generator [link]

Tech: *Next.js 15, React 19, TypeScript, GPT-4.1, OpenAI API, Inngest, E2B, PostgreSQL, Prisma, tRPC, Tailwind CSS, shadcn/ui*

- Developed an AI-powered code generation platform that converts natural language prompts into fully functional Next.js applications, reducing application prototyping time by 80%+.
- Built an autonomous AI agent pipeline using GPT-4.1, Inngest, E2B Sandboxes, TypeScript, PostgreSQL, Prisma, and tRPC to generate, deploy, and preview applications in real time.

- Engineered a cloud-based execution environment supporting live code generation, dependency installation, hot-reload previews, and source code visualization, enabling end-to-end application creation within minutes.

CrossCamReID – Multi-Camera Person Re-Identification Pipeline [link]

Tech: *ResNET, TensorRT, Embedding, ChromaDB, ByteTrack/BotSORT, FastAPI, WebSocket, Docker, YAML*

- Developed a real-time distributed analytics application for multi-camera people tracking; engineered scalable identity association workflows with ByteTrack/BotSORT, global identity memory, and dwell-time analytics — applying microservices design and modular inference pipeline architecture.
- Built session-aware FastAPI and WebSocket backend for low-latency streaming, occupancy monitoring, and threshold-based alerting; containerized using Docker with YAML-driven runtime configuration and CI/CD-ready deployment structure for GPU edge environments.

HACKATHONS

NVIDIA Nemotron Model Reasoning Challenge *Kaggle, 2026*

- Improved reasoning performance of NVIDIA Nemotron-3-Nano-30B through prompt engineering, synthetic data generation, and LoRA-based fine-tuning.
- Designed evaluation pipelines and optimization strategies for large language model reasoning benchmarks using modern AI tooling and GPU infrastructure.

AI Agent Security – Multi-Step Tool Attacks *OpenAI × Google × IEEE | Kaggle, 2026*

- Designed automated attack discovery algorithms to identify security vulnerabilities in tool-using AI agents through multi-step interaction chains.
- Evaluated agent robustness against exfiltration, untrusted-to-action, destructive write, and confused deputy attack scenarios using offline benchmark environments.

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft AI/ML Developer Professional Certificate | Oracle Cloud Infrastructure Foundations | Python for Data Science & AI – Coursera | Intro to TensorFlow for Deep Learning – Udacity | Google Cloud Arcade Program | Introduction to Cybersecurity – Cisco

EDUCATION

B.Tech in Computer Science & Engineering *2021 – 2025*

IILM College of Engineering & Technology, Greater Noida | GPA: 7.28